

If you only
have
1 minute

Choose a model that matches the deformation properties—not simply one that uses a neural network. Inter-subject brain registration involves smaller, more locally varying deformations, whereas intra-subject lung registration involves larger, smoother, more coherent motion. Our controlled comparison shows that models behave differently across these tasks. Matching the model prior and optimization path to the deformation properties is key to registration accuracy. Classical multiresolution B-spline registration remains a strong baseline and can also be fast (≈ 10 s, or even ≈ 2 s, with a modern GPU-accelerated PyTorch implementation).

Introduction

Motivation

Continuous DIR methods impose different deformation priors through their parameterization and optimization. B-splines offer local control and an explicit spatial scale. Recent INR-based methods show different performance in intra-subject lung CT and inter-subject brain MRI.

Objective

Disentangle whether improvement comes from the **neural network**, the **B-spline parameterization**, or a better **prior–deformation match** through controlled comparisons.

Method

Continuous parameterizations

INR-Dense (IDIR) [1]: SIREN predicts displacement.

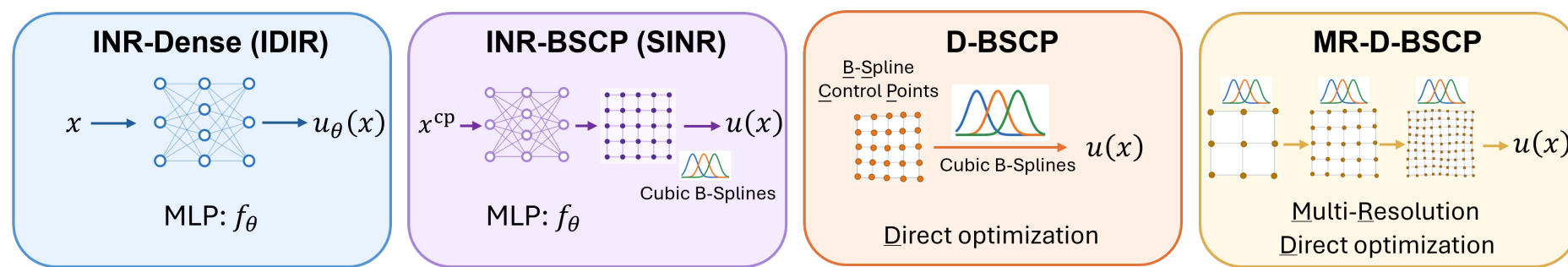
INR-BSCP (SINR) [2]: SIREN predicts B-spline control-point displacements.

D-BSCP: directly optimizes the control points.

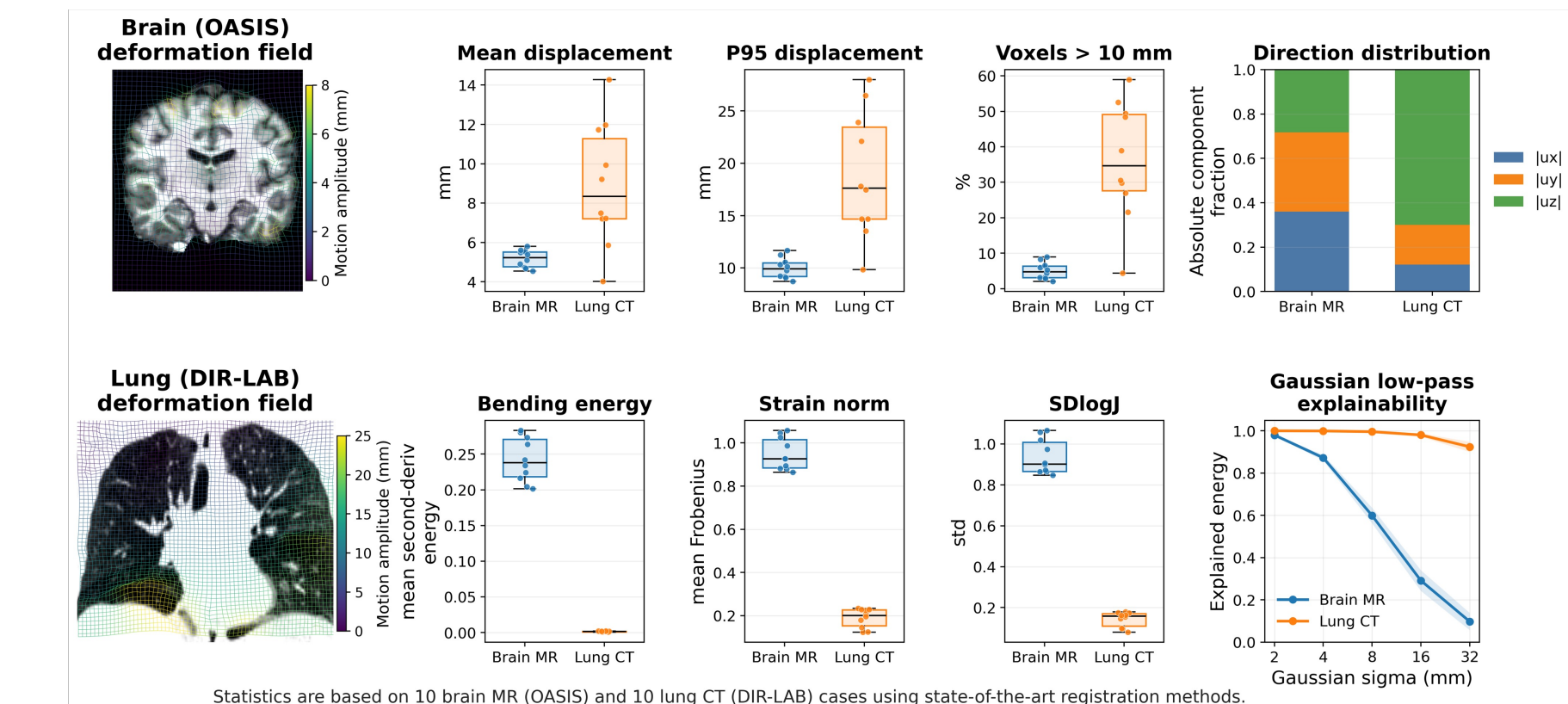
MR-D-BSCP: adds coarse-to-fine optimization.

For lung CT, we also evaluate **MR-INR-Dense (Dual-INR) [3]**, a multiscale variant.

A. Continuous Registration Parameterizations



B. Different Deformation Patterns



Off-grid is not enough: model priors should match deformation characteristics.

Fig. 1. Tested parameterizations and deformation patterns. Brain deformation is smaller and locally variable; lung motion is larger, smoother, and more coherent.

Implementation and optimization control

Official codebases: INR-Dense (IDIR) [1], INR-BSCP (SINR) [2], and MR-INR-Dense (Dual-INR) [3]. The first two retain their SIREN architectures (3 hidden layers $\times$ 256 units), sampling strategies, and similarity losses. MR-INR-Dense omits the lobe-segmentation loss. Direct spline variants are implemented in PyTorch.

Base budget: Adam, 2,500 steps per image pair, divided across levels for MR-D-BSCP. The same learning-rate sweep $\{10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}\}$ selects stable settings: 10^{-4} for INR; 10^{-3} for direct splines.

Hardware: NVIDIA RTX 6000 Ada, 48 GB.

Task-specific multiresolution schedules

Task	Control-point spacing (mm)	Image downsampling
OASIS	8 $\rightarrow$ 4 $\rightarrow$ 2	4 $\rightarrow$ 2 $\rightarrow$ 1
DIR-LAB 4DCT	32 $\rightarrow$ 16 $\rightarrow$ 8 $\rightarrow$ 4	8 $\rightarrow$ 4 $\rightarrow$ 2 $\rightarrow$ 1
DIR-LAB COPD	64 $\rightarrow$ 32 $\rightarrow$ 16 $\rightarrow$ 8 $\rightarrow$ 4	16 $\rightarrow$ 8 $\rightarrow$ 4 $\rightarrow$ 2 $\rightarrow$ 1

All image grids are 1 mm isotropic. The coarser DIR-LAB COPD start accommodates larger motion.

Evaluation

OASIS: 314/20/80 train/validation/test images; 100 test pairs; Dice/HD95 over 35 labels. **DIR-LAB 4DCT:** 10 cases; 300 landmark pairs per case; TRE on the original grid. **ARC curves [4]** sweep regularization strength; foreground **SD(log J)** measures regularity. Learning-based and Greedy references are fixed operating points.

Diagnostic: supervised representation capacity vs. unsupervised optimization

Fit external reference fields with supervised MSE: **SITReg for OASIS**; **pTVReg for DIR-LAB 4DCT** (10 pairs each). References are **not ground truth**. Compare with unsupervised registration at approximately matched SD(log J) to separate representability from optimization.

Alignment–regularity characteristics (ARC)

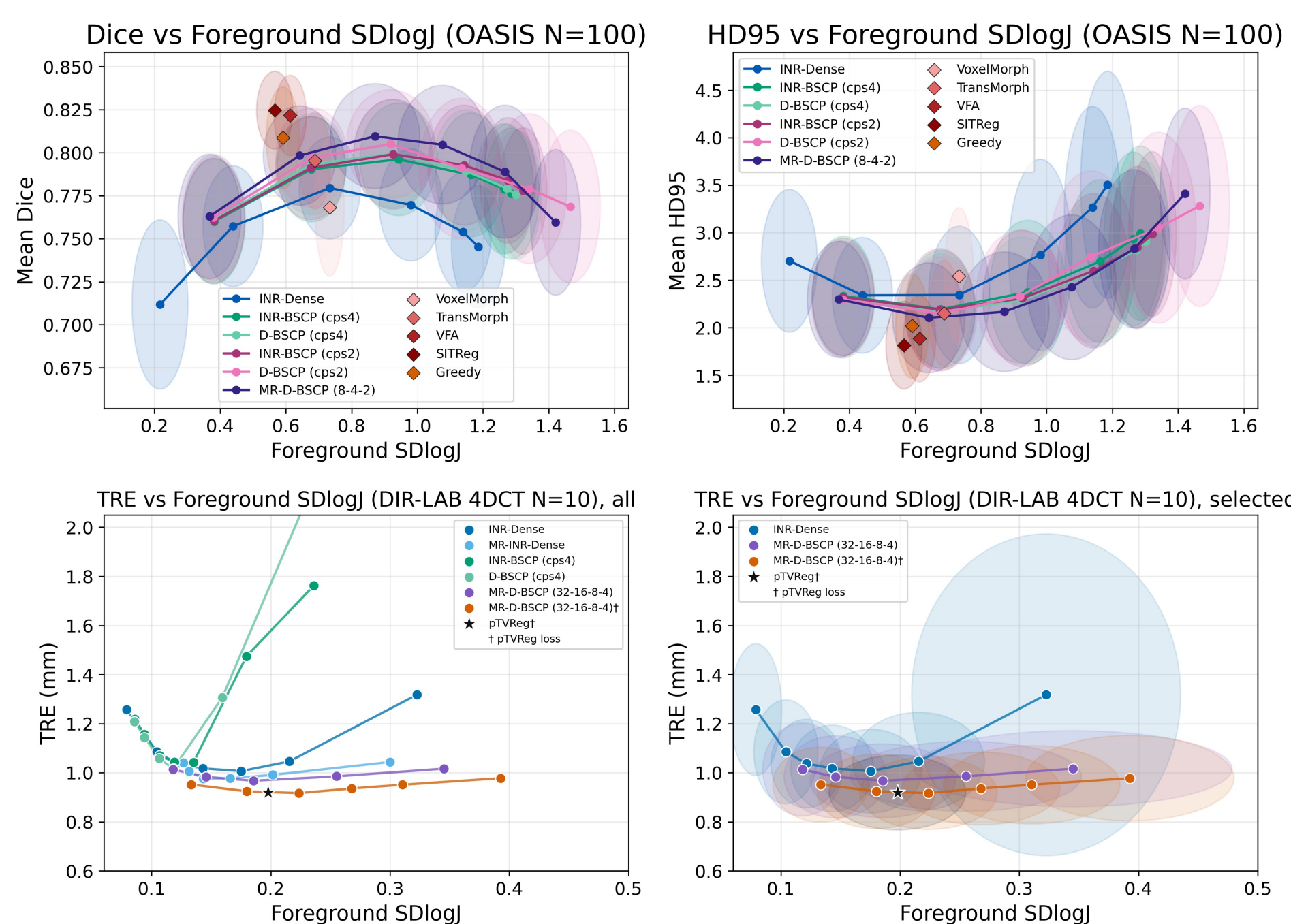


Fig. 2. ARC curves on OASIS (top) and DIR-LAB 4DCT (bottom). Ellipse axes indicate SD. ARC curves enable comparison across the full alignment–regularity spectrum, revealing the Pareto front between alignment and regularity.

OASIS: D-BSCP closely matches INR-BSCP at both control-point spacings. **MR-D-BSCP is best among the tested continuous parameterizations**; SITReg, VFA, and Greedy remain strong external references.

DIR-LAB 4DCT: single-scale spline methods become unstable at weak regularization. **INR-Dense is the only robust single-scale model**. Multiresolution helps both families, especially direct splines. The pTVReg isotropic-TV loss [5] improves lung registration over bending energy.

Case-wise results of DIR-LAB 4DCT & COPD

Table 1. Landmark TRE (mm; lower is better) on DIR-LAB 4DCT (N=10) and DIR-LAB COPD (N=10). Each case has 300 landmark pairs.

DIR-LAB 4DCT			DIR-LAB COPD		
Case	pTVReg	MR-D-BSCP	Case	pTVReg	MR-D-BSCP
1	0.7671	0.7920	1	0.7130	0.7343
2	0.7509	0.7804	2	<u>1.8987</u>	<u>3.1162</u>
3	0.9274	0.9401	3	0.7744	0.8407
4	1.2657	1.2501	4	0.6754	0.7986
5	1.0777	1.0934	5	0.7131	0.7671
6	0.8316	0.8270	6	0.6622	0.9279
7	0.8061	0.7974	7	0.7542	0.7997
8	1.0142	0.9975	8	0.7810	0.9721
9	0.9095	0.8849	9	0.6390	0.6253
10	0.8410	0.8053	10	0.8498	0.9190
Mean	0.9191	0.9168	Mean	0.8461	1.0501
SD	0.1607	0.1561	SD	0.3752	0.7331

MR-D-BSCP closely matches pTVReg’s TRE on all cases except COPD #2.

pTVReg [5] reference: updated GitHub benchmark (closest published match). Both: 1-mm isotropic; pyramid factor 0.7. DIR-LAB 4DCT: 4-mm control grid, no refinement, approximate LCC. DIR-LAB COPD: 8 $\rightarrow$ 4-mm control grid refinement, non-approximate LCC and cubic image interpolation.

Acceleration and practical speed

Table 2. OASIS accuracy, optimization steps, runtime, and peak GPU memory: base $\rightarrow$ fast.

Method	Dice base $\rightarrow$ fast	Steps base $\rightarrow$ fast	Cold (s) base $\rightarrow$ fast	Warm (s) base $\rightarrow$ fast	Peak GPU (GiB)
MR-D-BSCP (8-4-2)	0.810 $\rightarrow$ 0.809	2500 $\rightarrow$ 438	74.32 $\rightarrow$ 21.62	57.72 $\rightarrow$ 9.15	1.93
D-BSCP (cps2)	0.805 $\rightarrow$ 0.805	2500 $\rightarrow$ 400	91.92 $\rightarrow$ 22.64	81.55 $\rightarrow$ 13.14	1.84
D-BSCP (cps4)	0.799 $\rightarrow$ 0.797	2500 $\rightarrow$ 477	93.17 $\rightarrow$ 25.30	82.35 $\rightarrow$ 15.86	1.50
INR-BSCP (cps2)	0.799 $\rightarrow$ 0.793	2500 $\rightarrow$ 821	284.79 $\rightarrow$ 116.65	273.49 $\rightarrow$ 89.49	7.63
INR-BSCP (cps4)	0.796 $\rightarrow$ 0.786	2500 $\rightarrow$ 671	121.17 $\rightarrow$ 39.66	110.70 $\rightarrow$ 29.94	2.25

Image size: 160 $\times$ 192 $\times$ 224 voxels 1 $\times$ 1 $\times$ 1 mm³ isotropic.

Base: fixed learning rate and 2,500 steps.

Fast: stage-adaptive learning rates and convergence-based early stopping.

Cold: fresh process, including one-time CUDA/kernel initialization.

Warm: subsequent registrations in the same process.

Timing runs from input images to saved dense DVF; evaluation metrics are excluded.

MR-D-BSCP: 57.72 $\rightarrow$ 9.15 s warm runtime, with Dice nearly unchanged.

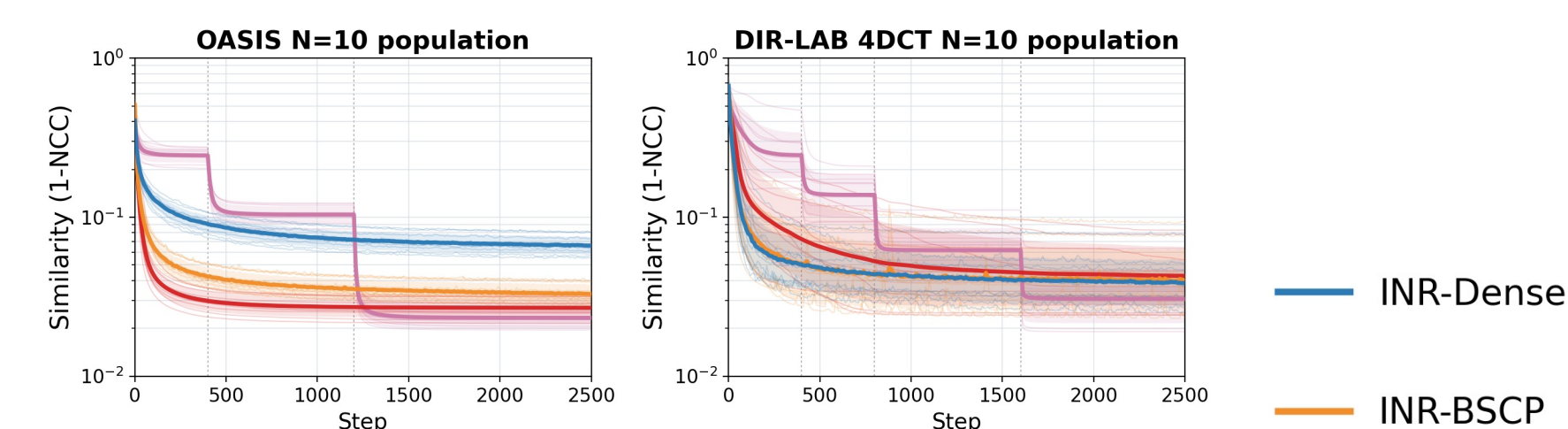
Further optimization shows that a warm runtime of ≈ 2 s is possible.

Results

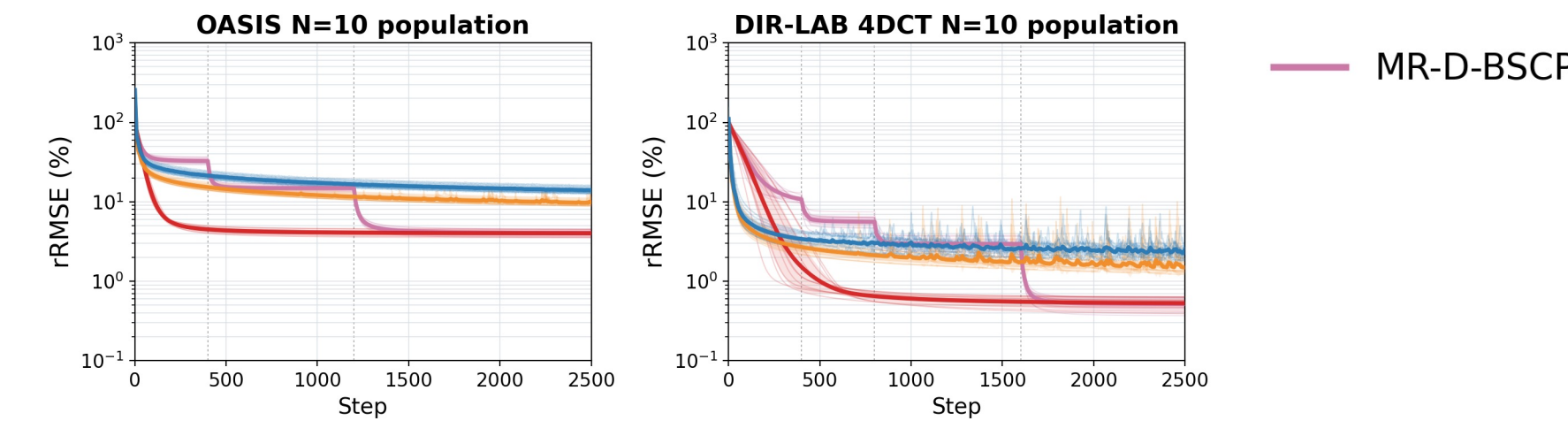
Diagnostic: representation capacity vs. optimization

Population-level behavior

A. Similarity metric (1-NCC) during unsupervised registration optimization

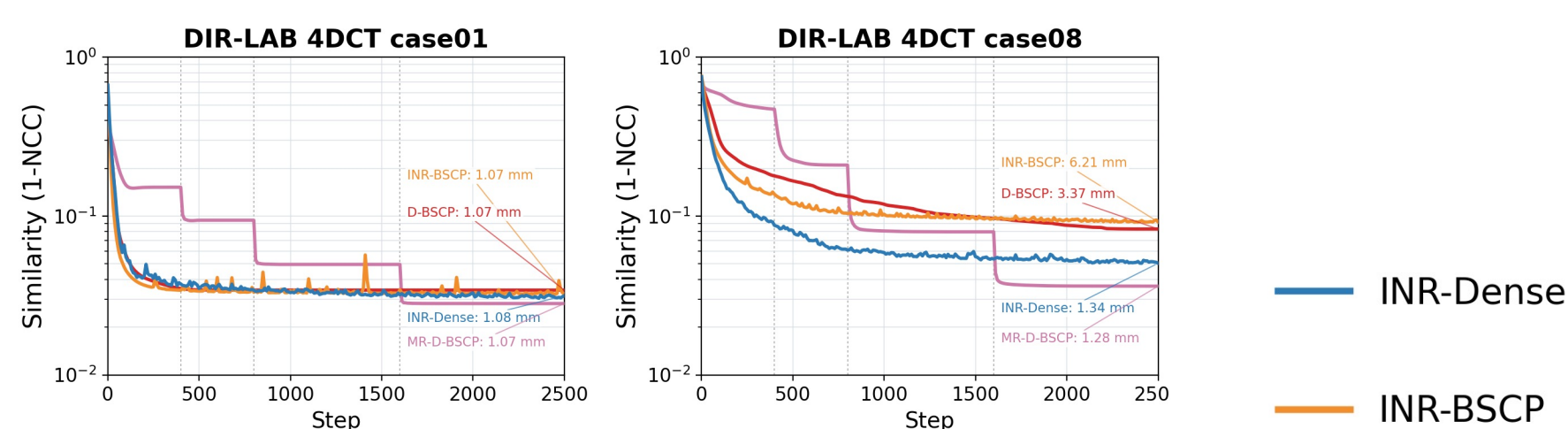


B. Target field fitting error (rRMSE) during supervised fitting



Representative DIR-LAB 4DCT cases

A. Similarity metric (1-NCC) during unsupervised registration optimization



B. Target field fitting error (rRMSE) during supervised fitting

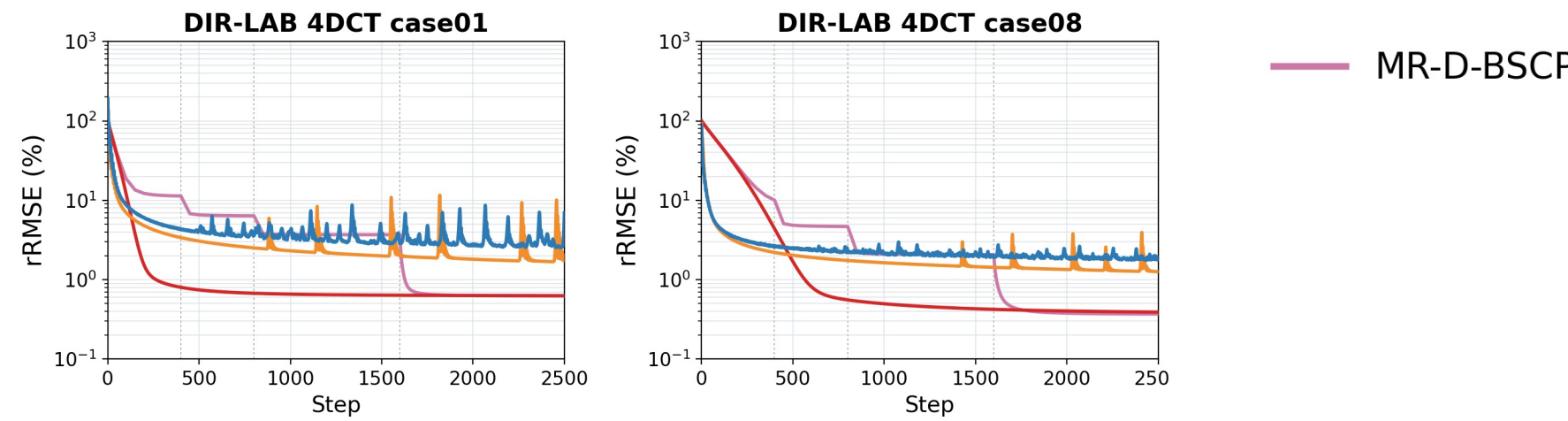


Fig. 3. Top block: population results (10 OASIS pairs; 10 DIR-LAB 4DCT cases). Bottom block: DIR-LAB 4DCT cases 01 and 08. Each block compares unsupervised registration (A) with supervised reference-field fitting (B).

OASIS: direct spline methods fit the reference fields well and also optimize well during registration. **DIR-LAB:** this relationship is less consistent across cases.

Good target fitting does not guarantee good registration. In large-motion case 08, D-BSCP fits the reference field well but fails in unsupervised registration. INR-Dense fits less accurately yet registers better. Multiresolution allows MR-D-BSCP to recover a good solution.

Conclusion and Discussion

1. Match the prior and optimization path.

Choose a parameterization and optimization strategy that fit the target deformation—not merely a trendy model.

2. INR is useful, not automatically superior.

Separate the neural contribution from the spline parameterization and coarse-to-fine refinement.

3. Retain a strong classical baseline.

Multiresolution B-spline optimization combines high registration accuracy with competitive speed.

4. Validate what helps and why more rigorously.

Use controlled ablations to investigate the mechanisms of improvement. ARC curves are preferred than selected single operating point.

Limitations: the effects of regularization form were not fully isolated (only bending energy and isotropic-TV tested); SIREN-specific hyperparameters (e.g., ω_0) were not retuned; broader task validation is needed.

Connection to learning-based DIR [6,7]

Recent studies emphasize **registration-specific designs** (e.g., *multiresolution*, *inverse consistency*, and *correlation computation*), rather than simply adopting a trendy network backbone. Our results extend this message to continuous pair-specific registration: **identify which components drive improvement and why, and match the design to the task.**

Off-Grid:

1st Workshop on Continuous Representations and Grid-Free Methods in Medical Imaging

References:

- [1] Wolterink JM, Zwienenberg JC, Brune C. Implicit Neural Representations for Deformable Image Registration. *MIDL, PMLR*. 2022.
- [2] Sideri-Lampretsa V, McGinnis J, Qiu H, et al. SINR: Spline-enhanced implicit neural representation for multi-modal registration. *MIDL, PMLR*. 2024.
- [3] Gebauer JB, Nielsen M, Madesta F, Werner R, Sentker T. Robust Multi-Scale Implicit Neural Representations for Large-Deformation Lung Registration. *MIDL, PMLR*. 2026.
- [4] Sideri-Lampretsa V, Rueckert D, Qiu H. Evaluation of Deformable Image Registration under Alignment-Regularity Trade-off. *LNCS*. 2026.
- [5] Vishnevskiy V, Gass T, Szekely G, Tanner C, Goksel O. Isotropic Total Variation Regularization of Displacements in Parametric Image Registration. *IEEE Trans Med Imaging*. 2017.
- [6] Jian B, Pan J, Ghahremani M, et al. Mamba? Catch the Hype or Rethink What Really Helps for Image Registration. *Biomedical Image Registration (WBIR)*. 2024.
- [7] Liu H, Ruan D, Sheng K. Unsupervised Deformable Image Registration Revisited: Enhancing Performance with Registration-Specific Designs. *MIDL Short Papers*. 2025.

